

Technical Datasheet 4G Gateway with 8DO & 8AI

Product Overview: -

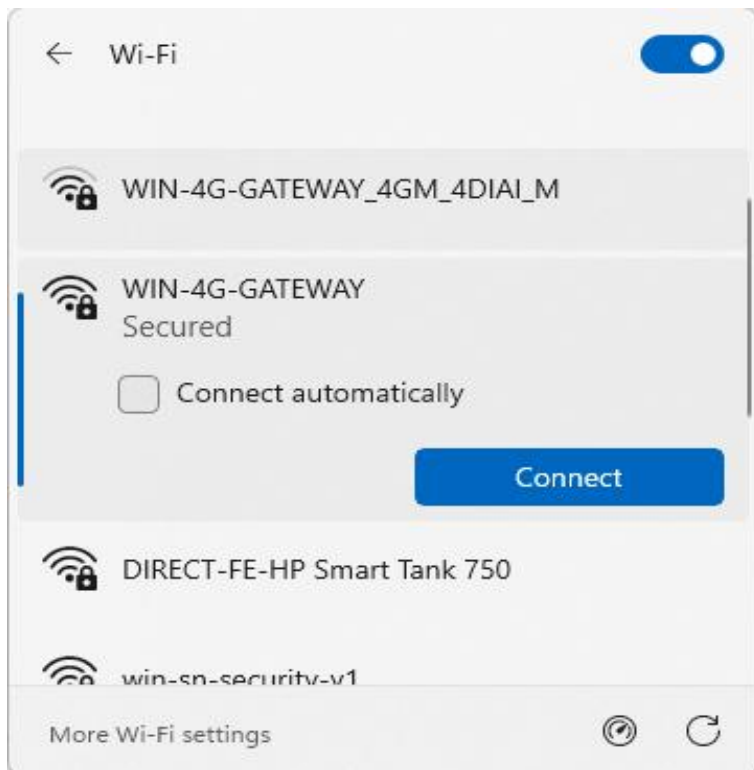
The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with an integrated I/O card featuring **8 Digital Outputs (DO)** and **8 Analog Inputs (AI)**, alongside a Modbus RS485 port and micro SIM compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



Specifications

General:-

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If 4G is not available, the system automatically switches to Wi-Fi. If both 4G and Wi-Fi are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Integrated I/O Channels:** Features 8 Digital Outputs (DO) for localized control and 8 Analog Inputs (AI) for direct sensor monitoring, expanding the gateway's standalone automation capabilities.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.



Configuration Settings: -

- Power up the module by connecting a 12-24 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: Wittelb-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage will be accessible.

Page1:-

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	modbus data	Logical Operations
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WiFi Configuration

SSID:

Password:

Device ID:

Data Frequency:

SD Data Upload Time:

Submit

- Enter Device ID (Should be unique for every device).
- Enter data frequency in min to receive data on cloud.
- Enter time to upload SD card data to cloud.

Page2:- On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical oprations
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Rs485 Configuration

Baud Rate:

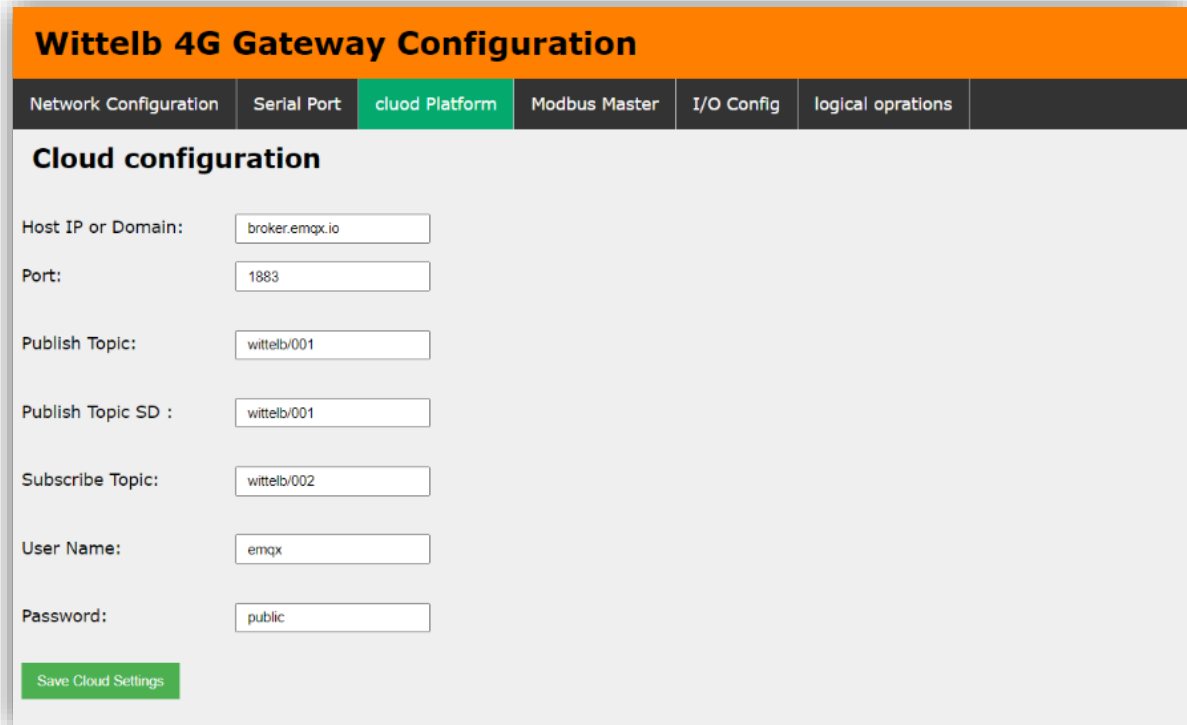
Data bits:

Parity:

Stop Bits:

Save RS485 Configuration

Page3:-



Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical oprations
Cloud configuration					
Host IP or Domain:		broker.emqx.io			
Port:		1883			
Publish Topic:		wittelb/001			
Publish Topic SD :		wittelb/001			
Subscribe Topic:		wittelb/002			
User Name:		emqx			
Password:		public			
Save Cloud Settings					

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.

Page4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- Indicate the register end address for the slave.
- Enter the function code corresponding to the desired slave.

Wittelb 4G Gateway Configuration

 Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | I/O Config | logical operations | About

Name	Slave Address	Register start	Quantity	Function Code	Actions
MFM	5	0	50	3	Delete
Add Row Submit					

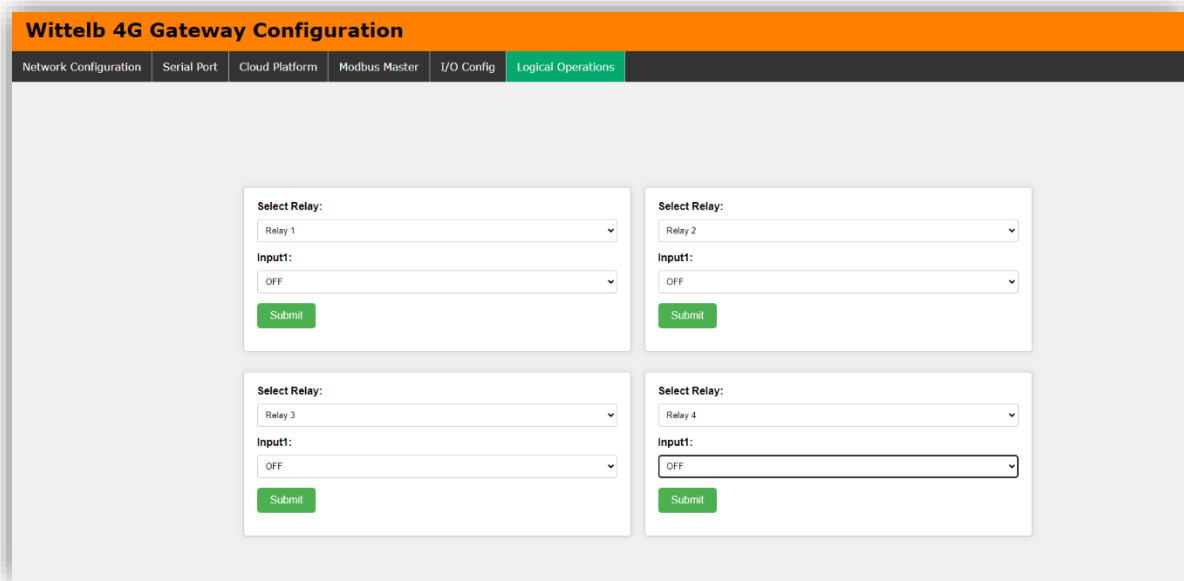
Page5:-

Wittelb 4G Gateway Configuration

 Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | **I/O Config** | logical operations

Sl.No	Input/Output	Name	Value
1	DO1	RLY1	
2	DO2	RLY2	
3	DO3	RLY3	1
4	DO4	RLY4	
5	DI1	DIS1	1
6	DI2	DIS2	
7	DI3	DIS3	
8	DI4	DIS4	
9	AI1	AI10-0	259
10	AI2	AI10-1	197
11	AI3	AI10-2	213
12	AI4	AI10-3	227
13	AI1	AI12-4	552
14	AI2	AI12-5	2
15	AI3	AI12-6	
16	AI4	AI12-7	1

- Go to page 5 on the I/O Config interface.
- You can view real-time values of both analog and digital inputs on the baseboard.
- Additionally, you have the option to assign specific names to each input/output (I/O) as needed.



Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | I/O Config | **Logical Operations**

Relay 1 Configuration:

Select Relay: Relay 1
Input1: OFF
Submit

Relay 2 Configuration:

Select Relay: Relay 2
Input1: OFF
Submit

Relay 3 Configuration:

Select Relay: Relay 3
Input1: OFF
Submit

Relay 4 Configuration:

Select Relay: Relay 4
Input1: OFF
Submit

- Navigate to page 6 on the Logical Operations interface.
- Here, you can configure logical AND and OR operations for each of the 4 relays.
- You have the flexibility to assign any two digital inputs to a relay and choose the desired logical operation.
- Additionally, you can assign an analog input to a relay and define minimum and maximum threshold values for relay operation.
- If you prefer not to use any logical operation, simply select the "OFF" option for input one.

Select Relay:

Relay 2

**Input1:**

DI3

**Logical
Operation:**

AND

**Input1 State:**

High

**Input2:**

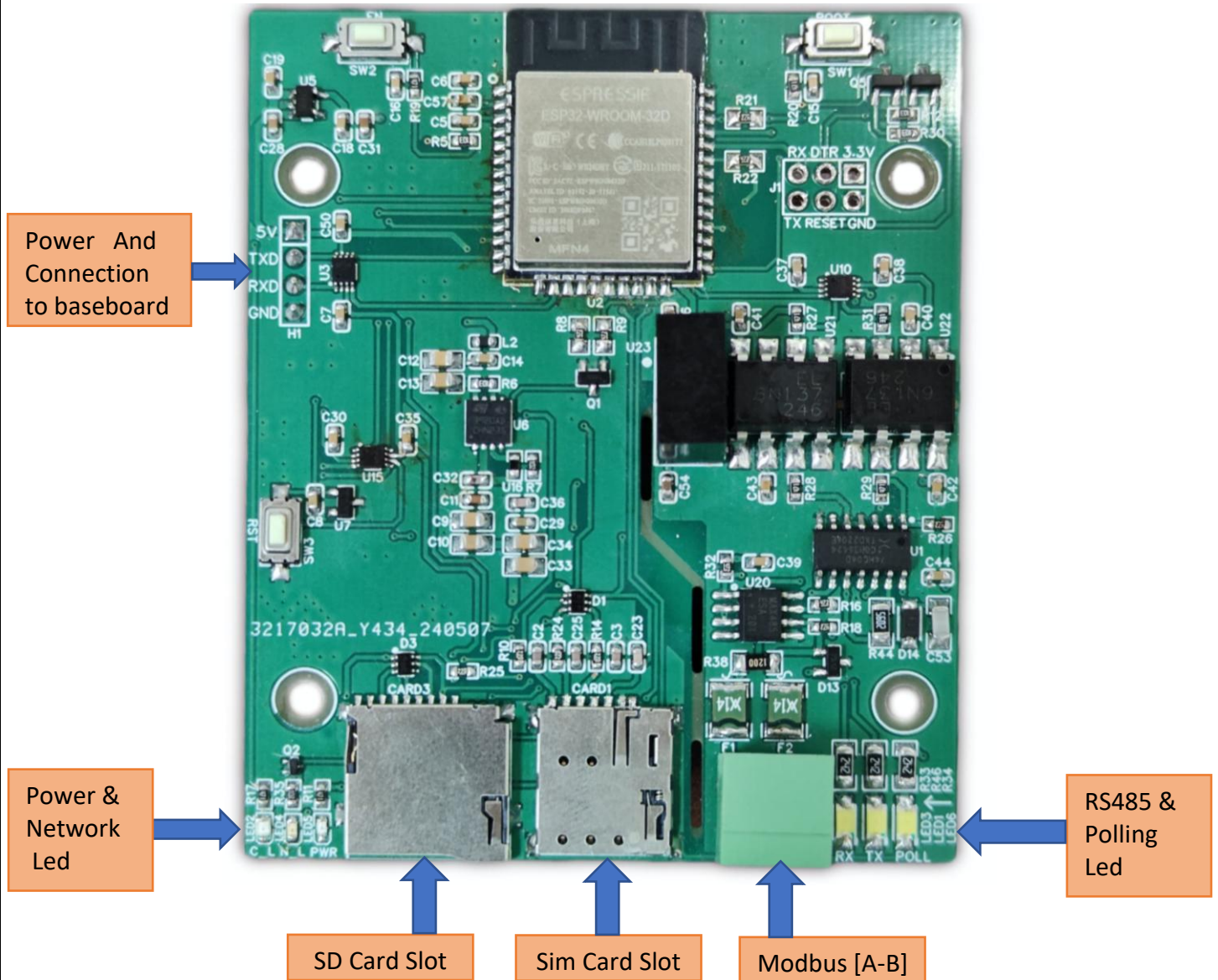
DI1

**Input2 State:**

High

**Submit**

4G Gateway connection Diagram:



Additional Features: -

Communication ports are isolated

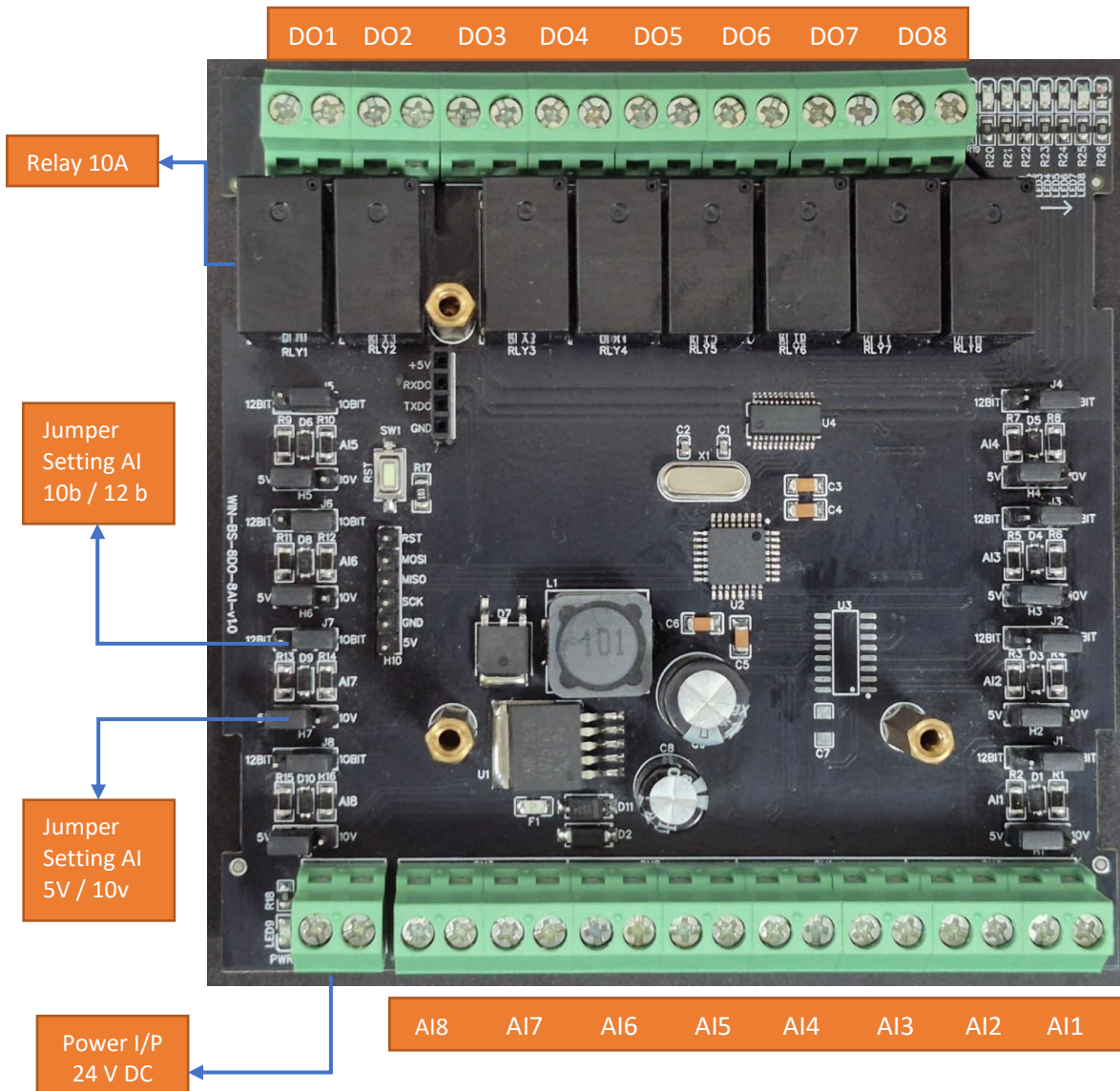
Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

400V isolation.

Base Board with 8 AI port with Jumper settings for 10 bit / 12 bit and 5v / 10 v Input Sources.



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